



AcademyCloudFoundatio...
PDF

Make notes Very detailed (full sentences, no short phrases)

- ✓ Including all important theoretical points
- ✓ Including procedural steps wherever steps are given
- ✓ Formatted cleanly for exam

make sure to include each and every imp point



Here are **very detailed, exam-ready notes** for **AWS Cloud Security (Module 4)**.
I have covered **every important concept, full explanations, and all procedural steps** clearly.



AWS Cloud Security – Detailed Notes

(From your PDF AcademyCloudFoundations_Module_...)

1. AWS Shared Responsibility Model

Question: Explain the AWS Shared Responsibility Model.

The AWS Shared Responsibility Model defines how security responsibilities are divided between AWS and the customer.

AWS is responsible for **security of the cloud**, while the customer is responsible for **security in the cloud**.

AWS Responsibilities (Security of the Cloud)

AWS is responsible for protecting the infrastructure that runs all AWS services. This includes:

- AWS ensures **physical security of data centers** to prevent unauthorized access.
- AWS provides **controlled, need-based access** to infrastructure.
- AWS manages **hardware and software infrastructure**, including servers and storage systems.

- AWS handles **network infrastructure security**, including routing and switching.
- AWS performs **intrusion detection and prevention** at infrastructure level.
- AWS ensures **virtualization infrastructure security** and proper instance isolation.
- AWS manages **logging, auditing, and storage decommissioning processes**.

👉 This is confirmed in page 6 of the document  AcademyCloudFoundations_Module_... .

Customer Responsibilities (Security in the Cloud)

Customers are responsible for configuring and managing their own resources securely.

- The customer manages the **operating system of EC2 instances**, including updates and patching.
- The customer is responsible for **applications and their security**.
- The customer manages **user access, passwords, and roles**.
- The customer configures **security groups and firewalls**.
- The customer handles **network configuration and monitoring**.
- The customer manages **data encryption and protection**.
- The customer is responsible for **account management and permission settings**.

👉 As shown in page 7  AcademyCloudFoundations_Module_... .

Service Models and Responsibility

Infrastructure as a Service (IaaS)

- Customer has maximum control.
- Customer manages OS, applications, and configurations.
- Example: EC2, EBS, VPC.

Platform as a Service (PaaS)

- AWS manages infrastructure and OS.
- Customer focuses on code and data.
- Example: RDS, Elastic Beanstalk, Lambda.

Software as a Service (SaaS)

- AWS manages everything.
 - Customer only uses the application.
 - Example: AWS Shield, AWS Trusted Advisor.
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Key Takeaways

- AWS secures infrastructure.
 - Customer secures data, applications, and access.
 - Responsibility varies depending on service type.
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2. AWS Identity and Access Management (IAM)

Question: Explain IAM and its components.

IAM is a service used to **control access to AWS resources securely**.

Key Features of IAM

- IAM allows defining **who can access resources**.
- It enables **fine-grained access control**.
- IAM is **free to use within an AWS account**.
- A resource can be anything like an EC2 instance or S3 bucket.

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IAM Components

1. IAM User

An IAM user is a person or application that can log in to AWS.

2. IAM Group

A group is a collection of users with the same permissions.

3. IAM Policy

A policy is a JSON document that defines permissions.

4. IAM Role

A role provides temporary permissions and can be assumed by users or services.

Authentication Methods

Programmatic Access

- Uses Access Key ID and Secret Access Key.
- Used for CLI and SDK.

Console Access

- Uses username, password, and account ID.
 - Can include MFA.
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Multi-Factor Authentication (MFA)

- MFA adds an extra layer of security.
 - Requires username + password + authentication code.
 - Prevents unauthorized access.
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Authorization in IAM

- By default, **all access is denied**.
- Explicit allow → grants access.
- Explicit deny → overrides all allows.

👉 Explained in page 22–26  AcademyCloudFoundations_Module_...

IAM Policies

- Policies define **what actions are allowed or denied**.
 - Types:
 - Identity-based policies (attached to users, groups, roles)
 - Resource-based policies (attached to resources like S3)
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Principle of Least Privilege

Users should only be given **minimum permissions required** to perform their tasks.

IAM Groups

- Used to assign permissions to multiple users.
 - Users can belong to multiple groups.
 - Groups cannot be nested.
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IAM Roles

- Provide temporary access.
- Not tied to a specific user.

- Common use: EC2 accessing S3.
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Key Takeaways

- IAM controls authentication and authorization.
 - Policies define permissions.
 - Roles are used for temporary and secure access.
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3. Securing a New AWS Account

Question: Explain steps to secure a new AWS account.

Step 1: Stop Using Root User

The root user has full access and should not be used regularly.

Procedure:

1. Create an IAM user.
2. Create an admin group.
3. Add IAM user to group.
4. Disable root access keys.
5. Enable password policy.
6. Store root credentials securely.



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Step 2: Enable MFA

- Enable MFA for root and all IAM users.
 - Use apps like Google Authenticator or hardware tokens.
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Step 3: Enable AWS CloudTrail

CloudTrail records all API activities.

Steps:

1. Open CloudTrail in console.
2. View event history.

3. Create a trail for long-term logging.
 4. Store logs in S3 bucket.
 5. Restrict access to logs.
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Step 4: Enable Billing Reports

- Helps monitor AWS usage and costs.
 - Reports are stored in S3.
 - Updated daily.
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Best Practices

- Use MFA.
 - Do not use root user regularly.
 - Use IAM users and groups.
 - Apply least privilege.
 - Monitor activity with CloudTrail.
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4. Securing AWS Accounts

AWS Organizations

- Used to manage multiple AWS accounts.
 - Accounts can be grouped into Organizational Units (OUs).
 - Policies can be applied centrally.
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Service Control Policies (SCPs)

- Define maximum permissions.
 - Do not grant permissions directly.
 - Ensure compliance across accounts.
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AWS Key Management Service (KMS)

- Used to create and manage encryption keys.
 - Uses secure hardware modules (HSMs).
 - Integrated with CloudTrail.
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Amazon Cognito

- Provides authentication for apps.
 - Supports social logins (Google, Facebook).
 - Scales to millions of users.
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AWS Shield

- Protects against DDoS attacks.
 - Provides automatic detection and mitigation.
 - Comes in Standard (free) and Advanced (paid).
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5. Securing Data on AWS

Encryption of Data at Rest

- Data stored on disk is encrypted.
- Requires a secret key.
- Managed using AWS KMS.

Supported services:

- S3
 - EBS
 - EFS
 - RDS
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Encryption of Data in Transit

- Data moving over network is encrypted.
 - Uses TLS/SSL protocols.
 - HTTPS ensures secure communication.
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Securing S3 Buckets

- Buckets are private by default.
- Access must be controlled.

Methods:

- IAM policies
- Bucket policies

- ACLs (legacy)
 - Block Public Access feature
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6. Compliance in AWS

AWS Compliance Programs

AWS supports compliance through:

Certifications

- ISO 27001, ISO 27017, ISO 27018

Laws & Regulations

- GDPR
- HIPAA

Frameworks

- CIS benchmarks
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AWS Config

- Monitors and evaluates configurations.
 - Tracks changes over time.
 - Helps in auditing and compliance.
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AWS Artifact

- Provides compliance reports.
 - Includes ISO, PCI, SOC reports.
 - Used for auditing and verification.
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Key Takeaways

- AWS provides tools for compliance monitoring.
 - Customers must ensure their own compliance.
 - AWS Artifact and Config help in auditing.
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- AWS follows a shared responsibility model.
 - IAM manages authentication and authorization.
 - Root user should be avoided and MFA must be enabled.
 - CloudTrail is used for monitoring activities.
 - Data must be encrypted both at rest and in transit.
 - AWS provides services like KMS, Cognito, and Shield for security.
 - Compliance is supported through AWS Config and Artifact.
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If you want, I can also:

- ✓ Convert this into **short revision notes (1–2 pages)**
- ✓ Give **important exam questions + answers**
- ✓ Or create **memory tricks for IAM & security topics**

Sources

